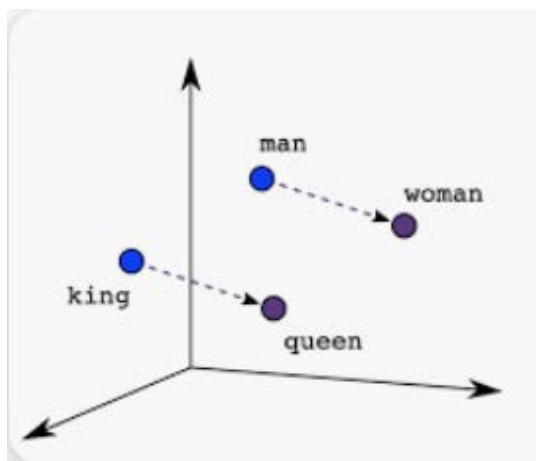
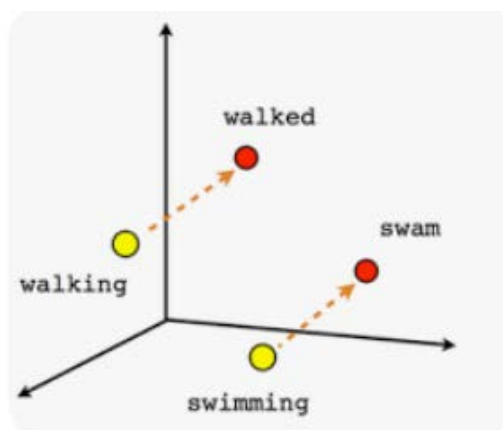


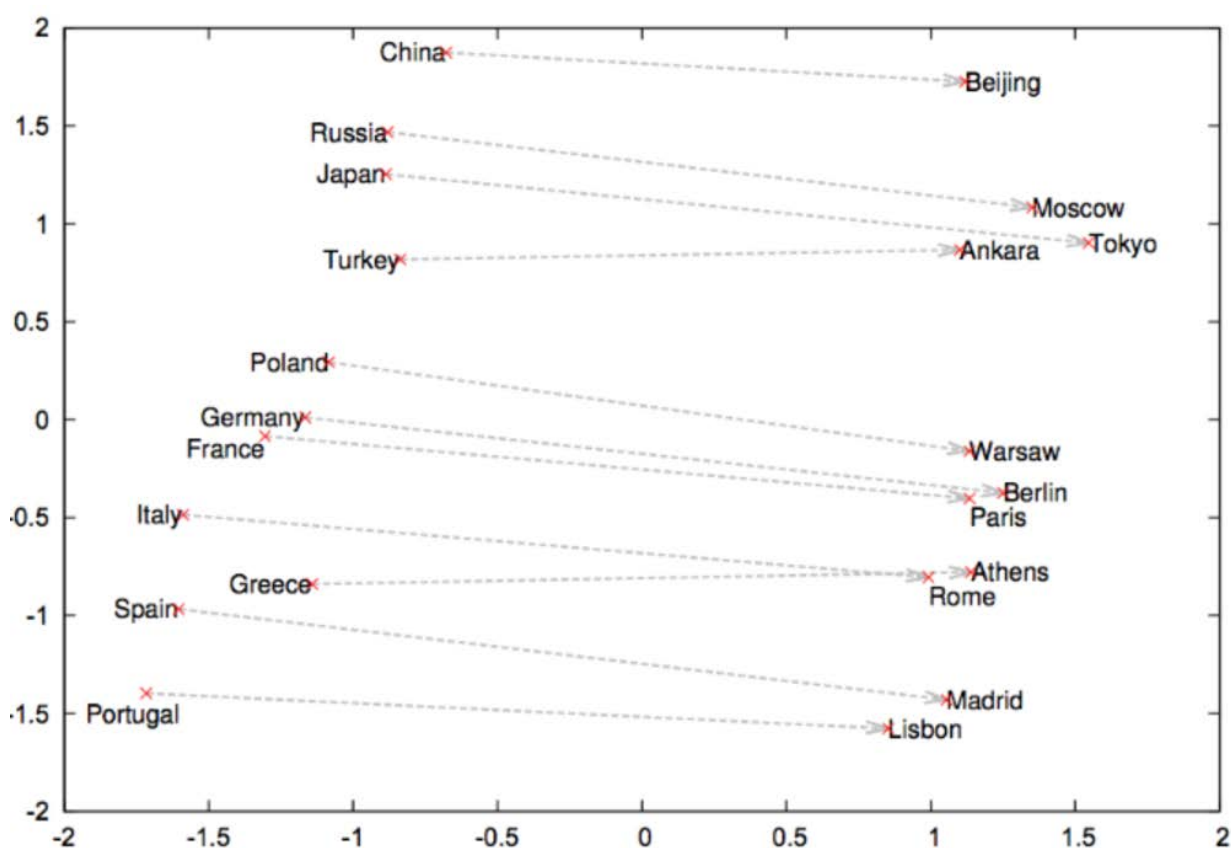
Analogy Reasoning Evaluation details



$$\text{Vec}(\text{king}) - \text{vec}(\text{queen}) = \text{vec}(\text{man}) - \text{vec}(\text{woman})$$



$$\text{Vec}(\text{walking}) - \text{vec}(\text{walked}) = \text{vec}(\text{man}) - \text{vec}(\text{woman})$$



$$\text{Vec}(\text{china}) - \text{vec}(\text{beijing}) = \text{vec}(\text{russia}) - \text{vec}(\text{moscow}) \quad [\text{Note: this relation is usually not exact}]$$

In **analogy reasoning** evaluation, we first compute the prediction vector:

$$\text{vec_pred} = \text{vec}(\text{china}) - \text{vec}(\text{beijing}) + \text{vec}(\text{moscow})$$

Then in the list of words, we search the word-embedding-vectors to find the closest word.

If the searched word is "russia", the analogy for these words grouping is **correct**. Otherwise, it's **false**..

We suggest to plot the two embedding vectors, **vec_pred**, **vec(russia)** in a figure.:

x-axis = dimension (total 64 or 128 dimensions), y-axis=value of each a vector in this dimension
It's better to use --- lines to connect all values in one vector, thus you have two lines, representing two vectors.